

RIGHT TRIANGLE TRIGONOMETRY

Full Solutions & Explanations

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Total Questions: 40

Q1. Correct Answer: **A • EASY**

STEP-BY-STEP SOLUTION:

Sine is defined as the ratio of the side opposite the angle to the hypotenuse: $\sin(\text{angle}) = \text{opposite/hypotenuse}$.

For angle A, the opposite side is BC (the side that does not touch angle A), and the hypotenuse is AB (opposite the right angle).

$$\sin(A) = BC/AB = 6/10 = 3/5.$$

WHY EACH OPTION IS WRONG:

A) Correct. $\sin(A) = \text{opposite/hypotenuse} = BC/AB = 6/10 = 3/5$.

B) This uses $AC/AB = 8/10 = 4/5$, which is $\cos(A)$, not $\sin(A)$ — AC touches angle A, making it the adjacent side, not the opposite one.

C) This uses $AC/BC = 8/6$ reduced to $4/3$, which mixes the two legs together instead of using opposite over hypotenuse.

D) This is the reciprocal of $\tan(A)$ (adjacent/opposite), not $\sin(A)$ at all.

TRAP ALERT

Confusing which side is "opposite" angle A — the opposite side never touches the angle, while AC, which DOES touch angle A, is the adjacent side, not the opposite one.

KEY INSIGHT

SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent. The opposite side never touches the angle in question; the adjacent leg does (besides the hypotenuse, which always touches the angle too).

FAST METHOD

Trace from the angle label: the one leg that does NOT touch that vertex is always the opposite side.

Q2. Correct Answer: **B • EASY**

STEP-BY-STEP SOLUTION:

Cosine is defined as the ratio of the side adjacent to the angle to the hypotenuse: $\cos(\text{angle}) = \text{adjacent/hypotenuse}$.

Since angle D is the right angle, EF is the hypotenuse (the side opposite the right angle).

The sides touching angle F are DF and EF; since EF is the hypotenuse, the adjacent leg is DF = 12.

$$\cos(F) = \text{adjacent/hypotenuse} = DF/EF = 12/13.$$

WHY EACH OPTION IS WRONG:

A) This uses $DE/EF = 5/13$, but DE is the side opposite angle F (it doesn't touch F), so this is $\sin(F)$, not $\cos(F)$.

- B)** Correct. $\cos(F) = \text{adjacent/hypotenuse} = DF/EF = 12/13$.
C) This uses $DF/DE = 12/5$, mixing the two legs together instead of comparing the adjacent leg to the hypotenuse.
D) This is the reciprocal of choice C and still doesn't involve the hypotenuse at all.

TRAP ALERT 🚧

Mixing up DE and DF — since both start with D, it's easy to lose track of which one actually touches angle F (the adjacent leg) versus which one is opposite F.

KEY INSIGHT 💡

When two legs share a letter with the right-angle vertex, identify adjacency by checking which leg's OTHER endpoint matches the angle you're evaluating.

FAST METHOD ⚡

The hypotenuse is always the side opposite the right-angle letter (here, EF, since D is the right angle). Once you know the hypotenuse, the adjacent leg is whichever remaining leg shares a letter with the angle in question.

Q3. Correct Answer: **C • EASY**

STEP-BY-STEP SOLUTION:

Tangent is defined as the ratio of the opposite side to the adjacent side: $\tan(\text{angle}) = \text{opposite/adjacent}$.

For angle H, the adjacent leg is GH = 9 (it touches H and is not the hypotenuse), and the opposite side is GI = 12 (it does not touch H).

$\tan(H) = \text{opposite/adjacent} = GI/GH = 12/9 = 4/3$.

WHY EACH OPTION IS WRONG:

- A)** This uses $GI/HI = 12/15 = 4/5$, which is $\sin(H)$, not $\tan(H)$ — it incorrectly uses the hypotenuse.
B) This uses $GH/HI = 9/15 = 3/5$, which is $\cos(H)$, again incorrectly involving the hypotenuse.
C) Correct. $\tan(H) = \text{opposite/adjacent} = GI/GH = 12/9 = 4/3$.
D) This flips the correct ratio, giving adjacent/opposite ($GH/GI = 9/12 = 3/4$) instead of opposite/adjacent.

TRAP ALERT 🚧

Accidentally pulling the hypotenuse (HI = 15) into the tangent ratio, when tangent is the only one of the three basic ratios that never involves the hypotenuse at all.

KEY INSIGHT 💡

Tangent is the odd one out: it compares the two legs directly (opposite/adjacent) and never uses the hypotenuse.

FAST METHOD ⚡

If a ratio choice includes the longest side (the hypotenuse) and the question asks for tangent, that choice is automatically wrong — eliminate it immediately.

Q4. Correct Answer: **D • EASY**

STEP-BY-STEP SOLUTION:

In a 45-45-90 triangle, the two legs are equal, and the hypotenuse equals a leg multiplied by $\sqrt{2}$. Here, the leg length is 9.

Hypotenuse = $9\sqrt{2}$.

WHY EACH OPTION IS WRONG:

- A) This just repeats the leg length without applying the $\sqrt{2}$ relationship at all.
- B) This doubles the leg length ($9 \times 2 = 18$), confusing the 45-45-90 relationship with simple doubling.
- C) This divides by $\sqrt{2}$ instead of multiplying, inverting the correct relationship.
- D) Correct. Hypotenuse = leg $\times \sqrt{2} = 9\sqrt{2}$.

TRAP ALERT 🚧

Mixing up the 45-45-90 ratio (leg : leg : leg $\sqrt{2}$) with the 30-60-90 ratio (short leg : short leg $\times\sqrt{3}$: short leg $\times 2$), leading to multiplying or dividing by the wrong factor.

KEY INSIGHT 💡

45-45-90 triangle side ratio: leg : leg : hypotenuse = 1 : 1 : $\sqrt{2}$. The hypotenuse is always (leg) $\sqrt{2}$.

FAST METHOD ⚡

Whenever both legs of a right triangle are equal, immediately multiply that leg length by $\sqrt{2}$ to get the hypotenuse — no other computation needed.

Q5. Correct Answer: **21** • EASY 🖊️ GRID-IN

STEP-BY-STEP SOLUTION:

Use the Pythagorean theorem: $(\text{leg}_1)^2 + (\text{leg}_2)^2 = (\text{hypotenuse})^2$.

$$20^2 + (\text{leg}_2)^2 = 29^2.$$

$$400 + (\text{leg}_2)^2 = 841.$$

$$(\text{leg}_2)^2 = 841 - 400 = 441.$$

$$\text{leg}_2 = \sqrt{441} = 21.$$

COMMON WRONG ANSWERS & WHY:

- Gridding 9 — this comes from subtracting $29 - 20 = 9$ directly, treating the sides as if they combined additively instead of through the Pythagorean relationship.
- Gridding 441 — this is the squared value of the missing leg, before taking the final square root.

TRAP ALERT 🚧

Treating the hypotenuse and a leg as if they relate by simple subtraction, when the Pythagorean theorem actually requires squaring, subtracting, and then taking a square root.

KEY INSIGHT 💡

For any right triangle, the Pythagorean theorem $a^2 + b^2 = c^2$ (with c as the hypotenuse) is the only reliable way to find a missing side when you know the other two.

FAST METHOD ⚡

Recognize the 20-21-29 Pythagorean triple directly once you see 20 and 29 together — these specific triples come up often enough to memorize.

Q6. Correct Answer: **A** • EASY

STEP-BY-STEP SOLUTION:

In a 30-60-90 triangle, the side ratios are: side opposite 30° (shortest) : side opposite 60° : side opposite 90° (hypotenuse) = 1 : $\sqrt{3}$: 2.

The hypotenuse is always twice the length of the side opposite the 30° angle.

$$\text{Hypotenuse} = 2 \times 7 = 14.$$

WHY EACH OPTION IS WRONG:

- A)** Correct. The hypotenuse is twice the shortest side: $2 \times 7 = 14$.
- B)** This is the length of the side opposite the 60° angle, not the hypotenuse — the question asks for a different side.
- C)** This halves the given side instead of doubling it.
- D)** This triples the given side, possibly from adding all three ratio numbers ($1+\sqrt{3}+2$ misapplied) instead of using only the doubling relationship.

TRAP ALERT

Mixing up which side the question is actually asking about — the side opposite 60° ($7\sqrt{3}$) is a tempting answer because it's the next number in the standard ratio, but the question specifically asks for the hypotenuse.

KEY INSIGHT

In a 30-60-90 triangle, the three sides always follow the pattern: short leg, short leg $\times \sqrt{3}$, short leg $\times 2$ (the hypotenuse) — memorize this ratio cold.

FAST METHOD

If you're given the side opposite 30° and asked for the hypotenuse, just double it. No other calculation needed.

Q7. Correct Answer: **B** • EASY

STEP-BY-STEP SOLUTION:

The co-function identity states that $\sin(\theta^\circ) = \cos(90^\circ - \theta^\circ)$ for any acute angle θ .

Here, $\sin(40^\circ) = \cos(90^\circ - 40^\circ) = \cos(50^\circ)$.

Comparing to $\cos(x^\circ)$, we get $x = 50$.

WHY EACH OPTION IS WRONG:

- A)** This assumes x must equal the original angle (40), ignoring that sine and cosine of the SAME angle are generally different values.
- B)** Correct. Since $\sin(\theta) = \cos(90 - \theta)$, $x = 90 - 40 = 50$.
- C)** This adds an arbitrary 20 to the given angle instead of subtracting it from 90.
- D)** This treats the complementary angle relationship as if x should just be 90 itself, ignoring the subtraction of 40.

TRAP ALERT

Forgetting that the co-function identity requires subtracting the given angle FROM 90, not just using 90 directly or reusing the original angle.

KEY INSIGHT

$\sin(\theta^\circ)$ always equals $\cos((90 - \theta)^\circ)$ for acute angles — the two angles are complementary, meaning they sum to exactly 90° .

FAST METHOD

Whenever you see $\sin$ of one angle set equal to $\cos$ of another, immediately add the two angle measures together — they must sum to 90.

Q8. Correct Answer: **C • EASY**

STEP-BY-STEP SOLUTION:

Use the Pythagorean theorem: $8^2 + (\text{leg})^2 = 17^2$.

$$64 + (\text{leg})^2 = 289.$$

$$(\text{leg})^2 = 289 - 64 = 225.$$

$$\text{leg} = \sqrt{225} = 15.$$

WHY EACH OPTION IS WRONG:

- A)** This subtracts the two given side lengths directly ($17 - 8 = 9$), which is not how the Pythagorean theorem works.
- B)** This adds the two given side lengths directly ($17 + 8 = 25$), again ignoring the actual Pythagorean relationship.
- C)** Correct. $\sqrt{(17^2 - 8^2)} = \sqrt{225} = 15$.
- D)** This adds the squares instead of subtracting them: $\sqrt{(17^2 + 8^2)} = \sqrt{353}$, which would be appropriate only if 17 and 8 were both legs, not a leg and the hypotenuse.

TRAP ALERT 🚧

Adding the squares instead of subtracting when the hypotenuse is one of the two given values — the formula only adds two legs together; when solving for a leg, you must subtract.

KEY INSIGHT 💡

When solving for a leg (not the hypotenuse), rearrange $a^2 + b^2 = c^2$ to $(\text{missing leg})^2 = c^2 - (\text{known leg})^2$ — subtraction, not addition.

FAST METHOD ⚡

Recognize the 8-15-17 Pythagorean triple on sight once you see 8 and 17 paired together.

Q9. Correct Answer: **20/29 • EASY** 🖊️ **GRID-IN**

STEP-BY-STEP SOLUTION:

Sine is the ratio of the opposite side to the hypotenuse: $\sin(\text{angle}) = \text{opposite}/\text{hypotenuse}$.

The side of length 20 is given as the opposite side, and the hypotenuse is 29.

$$\sin(\text{angle}) = 20/29 \text{ (approximately } 0.690\text{)}.$$

COMMON WRONG ANSWERS & WHY:

- Gridding 21/29 — this uses the OTHER leg (21) as if it were the opposite side, instead of the specified side of length 20.
- Gridding 20/21 — this compares the two legs to each other (which would be tangent of the OTHER angle), instead of comparing the specified leg to the hypotenuse.

TRAP ALERT 🚧

With two legs given, it's easy to grab the wrong one when identifying which side is "opposite" — always match the exact side length the question specifies.

KEY INSIGHT 💡

Sine always compares a leg to the hypotenuse, never the two legs to each other (that relationship is tangent).

FAST METHOD ⚡

Place the specified side over the hypotenuse and you have your sine ratio immediately — no triangle-labeling required since the side is named directly.

Q10. Correct Answer: **D • EASY**

STEP-BY-STEP SOLUTION:

Tangent is the ratio of the opposite side to the adjacent side: $\tan(\text{angle}) = \text{opposite}/\text{adjacent}$.

Since angle F is the right angle, DE is the hypotenuse.

For angle D, the adjacent leg is DF = 12 (it touches D and is not the hypotenuse), and the opposite side is EF = 5 (it does not touch D).

$\tan(D) = \text{opposite}/\text{adjacent} = EF/DF = 5/12$.

WHY EACH OPTION IS WRONG:

- A)** This flips the correct ratio, giving adjacent/opposite ($DF/EF = 12/5$) instead of opposite/adjacent.
- B)** This uses $DF/DE = 12/13$, which is $\cos(D)$, not $\tan(D)$ — it incorrectly involves the hypotenuse.
- C)** This uses $EF/DE = 5/13$, which is $\sin(D)$, again incorrectly involving the hypotenuse.
- D)** Correct. $\tan(D) = \text{opposite}/\text{adjacent} = EF/DF = 5/12$.

TRAP ALERT 🚧

Pulling the hypotenuse (DE = 13) into the answer when the question asks for tangent, which only ever compares the two legs.

KEY INSIGHT 💡

Before computing any ratio, identify the hypotenuse first (it's opposite the right angle), then everything else falls into place: tangent uses only the two remaining legs.

FAST METHOD ⚡

Whenever you see the hypotenuse's length appear in a tangent answer choice, that choice can be eliminated immediately.

Q11. Correct Answer: **A • MEDIUM**

STEP-BY-STEP SOLUTION:

Since the right angle is at R, angles P and Q are the two acute angles of the triangle, and they must sum to 90° (complementary).

The co-function identity states that for complementary angles, $\sin(P) = \cos(Q)$.

Therefore, $\cos(Q) = \sin(P) = 5/13$, with no further calculation needed.

WHY EACH OPTION IS WRONG:

- A)** Correct. Since P and Q are complementary, $\cos(Q) = \sin(P) = 5/13$ directly.
- B)** This computes the third side of the triangle ($\sqrt{13^2 - 5^2} = 12$) and reports $12/13$, which is actually $\cos(P)$ or $\sin(Q)$ — unnecessary work that lands on the wrong angle's ratio.
- C)** This uses $5/12$, the tangent-style ratio of the two legs, instead of the requested cosine ratio.
- D)** This is the reciprocal of $\sin(P)$, not related to $\cos(Q)$ at all.

TRAP ALERT 🚧

Assuming you need to find the third side of the triangle and recompute everything from scratch, when the co-function identity makes that work completely unnecessary.

KEY INSIGHT 💡

Whenever two angles in a right triangle are complementary (sum to 90°), $\sin$ of one always equals $\cos$ of the other — no triangle measurements required.

FAST METHOD ⚡

See "sin" of one complementary angle and "cos" of the other in the same problem? They're equal. Write down the given value immediately as your answer.

Q12. Correct Answer: **B** • MEDIUM**STEP-BY-STEP SOLUTION:**

Since the right angle is at Z, angles X and Y are complementary (they sum to 90°).
The co-function identity states that for complementary angles, $\cos(X) = \sin(Y)$.
Therefore, $\sin(Y) = \cos(X) = 3\sqrt{5}/8$, with no further calculation needed.

WHY EACH OPTION IS WRONG:

- A)** This drops the radical entirely, reporting only the rational part of the given value.
- B)** Correct. Since X and Y are complementary, $\sin(Y) = \cos(X) = 3\sqrt{5}/8$ directly.
- C)** This takes the reciprocal of the given value instead of recognizing it should simply be copied over.
- D)** This swaps which number gets the radical sign, turning $3\sqrt{5}$ into $5\sqrt{3}$ — a careless transcription of the given expression.

TRAP ALERT 🚧

Even with an unfamiliar-looking radical expression, the co-function shortcut still applies exactly the same way — some students abandon the shortcut and try to build a whole triangle because the number looks more complicated than a simple fraction.

KEY INSIGHT 💡

The co-function identity $\sin(\theta) = \cos(90^\circ - \theta)$ works for any value, including expressions with radicals — the identity doesn't care how messy the number looks.

FAST METHOD ⚡

Don't let an unfamiliar radical expression tempt you into unnecessary computation — if the angles are complementary and the function flips ($\sin \leftrightarrow \cos$), just copy the value across.

Q13. Correct Answer: **51** • MEDIUM ✎ GRID-IN**STEP-BY-STEP SOLUTION:**

Since $\sin(\theta) = \text{opposite/hypotenuse} = 8/17$, this triangle is similar to (a scaled version of) an 8-15-17 right triangle.
The actual opposite side has length 24, so set up a proportion: $8/17 = 24/(\text{hypotenuse})$.
Cross-multiply: $8 \times \text{hypotenuse} = 17 \times 24 = 408$.
 $\text{hypotenuse} = 408/8 = 51$.

COMMON WRONG ANSWERS & WHY:

- Gridding 17 — simply gridding the denominator of the given ratio without scaling it to match the actual side length of 24.
- Gridding 45 — adding 17 and the scale factor ($24 - 8 = 16$, then $17+16=33$, or various other scaling slips) instead of correctly setting up and solving the proportion.

TRAP ALERT 🚧

Treating the given ratio $8/17$ as if it directly gave the hypotenuse length, forgetting that 8 and 17 only describe the RATIO of sides, not their actual measured lengths in this specific triangle.

KEY INSIGHT 💡

A trig ratio like $\sin(\theta) = 8/17$ describes the SHAPE of the triangle (any triangle similar to an 8-15-17 right triangle), not its exact size — use a proportion to scale up to the actual given side length.

FAST METHOD ⚡

Find the scale factor first: $24 \div 8 = 3$, then apply that same factor to the ratio's hypotenuse: $17 \times 3 = 51$.

Q14. Correct Answer: **C** • MEDIUM**STEP-BY-STEP SOLUTION:**

The angle of elevation, the horizontal distance, and the tower's height form a right triangle, with the height opposite the angle and the horizontal distance adjacent to it.

$\tan(50^\circ) = \text{opposite/adjacent} = \text{height}/60$.

$\text{height} = 60 \times \tan(50^\circ)$.

Using a calculator, $\tan(50^\circ) \approx 1.19175$, so $\text{height} \approx 60 \times 1.19175 \approx 71.5$, which rounds to 72 feet.

WHY EACH OPTION IS WRONG:

A) This comes from using $\sin(50^\circ)$ instead of $\tan(50^\circ)$: $60 \times \sin(50^\circ) \approx 46$, incorrectly treating 60 as the hypotenuse instead of the adjacent side.

B) This comes from using $\cos(50^\circ)$ in a similarly incorrect setup.

C) Correct. $\text{height} = 60 \times \tan(50^\circ) \approx 71.5$, which rounds to 72 feet.

D) This divides by $\tan(50^\circ)$ instead of multiplying: $60/\tan(50^\circ) \approx 50.4$, then a further miscalculation inflates this — in general, dividing instead of multiplying gives a value in the wrong direction entirely.

TRAP ALERT ⚠️

Misidentifying which sides the angle of elevation connects — the horizontal distance to the base is ADJACENT to the angle, and the height is OPPOSITE, which means tangent (not sine or cosine) is the correct ratio to use.

KEY INSIGHT 💡

For angle-of-elevation problems measured from a point on the ground, the horizontal distance is always adjacent, and the vertical height is always opposite — use tangent.

FAST METHOD ⚡

Set up $\text{height} = (\text{horizontal distance}) \times \tan(\text{angle})$ as your default formula for ground-level angle-of-elevation problems, then just plug in and compute.

Q15. Correct Answer: **D** • MEDIUM**STEP-BY-STEP SOLUTION:**

The angle of depression from the top of the lighthouse down to the boat equals the angle of elevation from the boat up to the top of the lighthouse (alternate interior angles).

This creates a right triangle where the lighthouse's height (120 feet) is opposite the 22° angle, and the distance to the boat is adjacent to it.

$\tan(22^\circ) = \text{opposite/adjacent} = 120/\text{distance}$, so $\text{distance} = 120/\tan(22^\circ)$.

Using a calculator, $\tan(22^\circ) \approx 0.4040$, so $\text{distance} \approx 120/0.4040 \approx 297.0$, which rounds to 297 feet.

WHY EACH OPTION IS WRONG:

- A)** This comes from multiplying instead of dividing: $120 \times \tan(22^\circ) \approx 48$, close to but not matching this exact wrong setup of using sine inappropriately.
- B)** This results from using $\sin(22^\circ)$ as if the lighthouse height were the hypotenuse: $120/\sin(22^\circ)$ is far too large, while $120 \times$ something near it produces this kind of smaller mismatched value from a different misapplied ratio.
- C)** This comes from a reversed or misapplied cosine setup instead of the correct tangent relationship.
- D)** Correct. distance = $120/\tan(22^\circ) \approx 297$ feet.

TRAP ALERT 🚨

Confusing angle of depression (measured downward from the horizontal at the TOP) with angle of elevation measured at the boat — they're equal in measure, but it's easy to set up the triangle with the wrong side as opposite versus adjacent.

KEY INSIGHT 💡

Angle of depression problems convert directly to angle of elevation problems using alternate interior angles — once converted, the known height is opposite and the unknown distance is adjacent, so divide by tangent.

FAST METHOD ⚡

For angle-of-depression-to-distance problems, the formula is distance = height/tan(angle) — memorize this as the standard setup.

Q16. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

The ladder, the wall, and the ground form a right triangle, with the ladder as the hypotenuse. Use the Pythagorean theorem: $10^2 + (\text{height})^2 = 26^2$.
 $100 + (\text{height})^2 = 676$.
 $(\text{height})^2 = 576$, so height = $\sqrt{576} = 24$ feet.

WHY EACH OPTION IS WRONG:

- A)** Correct. $\sqrt{(26^2 - 10^2)} = \sqrt{576} = 24$ feet.
- B)** This subtracts the two given lengths directly ($26 - 10 = 16$), ignoring the Pythagorean relationship entirely.
- C)** This adds the two given lengths directly ($26 + 10 = 36$), again ignoring the actual relationship between the sides.
- D)** This adds the squares instead of subtracting them ($\sqrt{(26^2 + 10^2)} \approx 27.9$), which would only apply if 10 and 26 were both legs rather than a leg and the hypotenuse.

TRAP ALERT 🚨

Adding the squares of the two given values instead of subtracting, which happens when a student forgets that the ladder length (26) is the hypotenuse, not a second leg.

KEY INSIGHT 💡

In any ladder-against-a-wall problem, the ladder itself is always the hypotenuse, the distance from the wall is one leg, and the height reached is the other leg.

FAST METHOD ⚡

Recognize the 10-24-26 Pythagorean triple (twice the 5-12-13 triple) immediately once you see 10 and 26 paired together.

Q17. Correct Answer: **180** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

The area of a right triangle is one-half the product of its two legs (the legs serve as base and height).

$$\text{Area} = (1/2)(9)(40).$$

$$\text{Area} = (1/2)(360) = 180.$$

COMMON WRONG ANSWERS & WHY:

- Gridding 360 — multiplying the two legs together but forgetting the factor of one-half required by the triangle area formula.
- Gridding 41 — mistakenly computing the hypotenuse (using the 9-40-41 Pythagorean triple) instead of the area that was actually asked for.

TRAP ALERT

Forgetting the $1/2$ factor in the triangle area formula, especially tempting here since $9 \times 40 = 360$ is itself a clean, satisfying-looking number that seems like it could be the final answer.

KEY INSIGHT

For a right triangle specifically, the two legs themselves serve as the base and height in the area formula, since they meet at a right angle — no need to find an altitude separately.

FAST METHOD

Multiply the two legs and immediately halve the result in one motion: $9 \times 40 = 360$, then $360 \div 2 = 180$.

Q18. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

Since $\sin(\theta) = \text{opposite/hypotenuse} = 7/25$, think of a right triangle with opposite side 7 and hypotenuse 25.

Find the adjacent side using the Pythagorean theorem: $(\text{adjacent})^2 + 7^2 = 25^2$, so $(\text{adjacent})^2 = 625 - 49 = 576$, giving $\text{adjacent} = 24$.

$$\cos(\theta) = \text{adjacent/hypotenuse} = 24/25.$$

WHY EACH OPTION IS WRONG:

- A)** This just repeats the given sine value instead of computing the actual cosine value.
- B)** Correct. The missing leg is 24 (from the 7-24-25 triple), so $\cos(\theta) = 24/25$.
- C)** This subtracts $25 - 7 = 18$ directly instead of applying the Pythagorean theorem correctly.
- D)** This compares the two legs to each other ($7/24$), which would be $\tan(\theta)$, not $\cos(\theta)$.

TRAP ALERT

Subtracting the given numbers directly ($25 - 7$) instead of using the Pythagorean theorem, which requires squaring first, then subtracting, then taking a square root.

KEY INSIGHT

Given any one trig ratio for an acute angle, you can find a right triangle's third side using the Pythagorean theorem, then build any of the other ratios directly from the three side lengths.

FAST METHOD

Recognize the 7-24-25 Pythagorean triple immediately once you see 7 and 25 together as legs and hypotenuse of a sine ratio.

Q19. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

Similar triangles have congruent corresponding angles — angle A and angle D have the exact same measure.

Since the angle measures are identical, all of their trigonometric ratios (sine, cosine, tangent) are also identical, regardless of how large or small each triangle actually is.

Therefore, $\sin(D) = \sin(A) = 5/13$.

WHY EACH OPTION IS WRONG:

- A)** This is the reciprocal of the given value, unrelated to how similarity actually works.
- B)** This introduces a different ratio (5/12) not present anywhere in the given information.
- C)** Correct. Since angle D has the same measure as angle A, $\sin(D) = \sin(A) = 5/13$.
- D)** This computes a different ratio (perhaps $\cos(A)$ using the 5-12-13 triple) instead of recognizing that $\sin(D)$ is simply equal to $\sin(A)$.

TRAP ALERT 🚧

Assuming that because triangle DEF might be a different size than triangle ABC, its trigonometric ratios must also be different — but trig ratios depend only on angle measure, never on the overall size of the triangle.

KEY INSIGHT 💡

Trigonometric ratios are properties of an ANGLE, not of a specific triangle's size. Similar triangles share all angle measures, so they share all trig ratios too.

FAST METHOD ⚡

Whenever a question mentions similar triangles and corresponding angles, the trig ratios transfer directly with no recalculation — just copy the given value across.

Q20. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Tangent can be found from sine and cosine using the identity $\tan(\theta) = \sin(\theta)/\cos(\theta)$.

$\tan(\theta) = (8/17)/(15/17)$.

Dividing by a fraction means multiplying by its reciprocal: $(8/17) \times (17/15) = 8/15$.

WHY EACH OPTION IS WRONG:

- A)** This is the reciprocal of the correct answer, possibly from flipping $\cos(\theta)/\sin(\theta)$ instead of $\sin(\theta)/\cos(\theta)$.
- B)** This incorrectly combines the two hypotenuse values (17 and 8) instead of properly dividing sine by cosine.
- C)** This incorrectly combines the two hypotenuse and adjacent values, not following the actual identity.
- D)** Correct. $\tan(\theta) = \sin(\theta)/\cos(\theta) = (8/17)/(15/17) = 8/15$.

TRAP ALERT 🚧

When dividing two fractions that share the same denominator (17 in both), it's easy to make an error keeping track of which numerators end up on top versus bottom in the final simplified ratio.

KEY INSIGHT 💡

The identity $\tan(\theta) = \sin(\theta)/\cos(\theta)$ holds for every angle — when sine and cosine share a denominator (as they do here, both over the hypotenuse), that denominator cancels out entirely, leaving just opposite/adjacent.

FAST METHOD ⚡

When $\sin(\theta)$ and $\cos(\theta)$ share the same denominator, ignore that denominator completely and just divide the two numerators directly: $8/15$.

Q21. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

The shadow's length and the flagpole's height form the two legs of a right triangle, with the shadow adjacent to the angle of elevation and the height opposite it.

$$\tan(50^\circ) = \text{opposite/adjacent} = \text{height}/18.$$

$$\text{height} = 18 \times \tan(50^\circ).$$

Using a calculator, $\tan(50^\circ) \approx 1.19175$, so $\text{height} \approx 18 \times 1.19175 \approx 21.45$, which rounds to 21 feet.

WHY EACH OPTION IS WRONG:

- A)** Correct. $\text{height} = 18 \times \tan(50^\circ) \approx 21.45$, which rounds to 21 feet.
- B)** This results from dividing by $\tan(50^\circ)$ instead of multiplying: $18/\tan(50^\circ) \approx 15.1$.
- C)** This comes from using $\sin(50^\circ)$ instead of $\tan(50^\circ)$: $18 \times \sin(50^\circ) \approx 13.8$, incorrectly treating 18 as the hypotenuse.
- D)** This comes from using $\cos(50^\circ)$ instead of $\tan(50^\circ)$: $18 \times \cos(50^\circ) \approx 11.6$, an even less appropriate setup.

TRAP ALERT 🚨

Using sine or cosine instead of tangent — since the shadow length (18) is a leg, not the hypotenuse, only tangent correctly relates it to the opposite leg (height).

KEY INSIGHT 💡

Shadow-and-height problems always form a right triangle where the shadow is adjacent to the elevation angle and the object's height is opposite — use tangent.

FAST METHOD ⚡

Multiply the shadow length by $\tan(\text{angle})$ to get the height directly: $\text{height} = \text{shadow} \times \tan(\text{angle})$.

Q22. Correct Answer: **11** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

In a 30-60-90 triangle, the hypotenuse is always twice the length of the side opposite the 30° angle.

$$\text{So the side opposite } 30^\circ = \text{hypotenuse}/2.$$

$$\text{Side opposite } 30^\circ = 22/2 = 11.$$

COMMON WRONG ANSWERS & WHY:

- Gridding 44 — doubling the hypotenuse instead of halving it, reversing the correct relationship.
- Gridding 19.05 (approximately) — mistakenly dividing the hypotenuse by $\sqrt{3}$ instead of by 2.

TRAP ALERT 🚨

Mixing up which side relates to the hypotenuse by a factor of 2 (the short leg, opposite 30°) versus by a factor of $\sqrt{3}$ (the long leg, opposite 60°).

KEY INSIGHT 💡

30-60-90 triangle ratio: short leg (opposite 30°) : long leg (opposite 60°) : hypotenuse = $1 : \sqrt{3} : 2$.
The hypotenuse is always twice the SHORT leg specifically.

FAST METHOD ⚡

Whenever you're given the hypotenuse of a 30-60-90 triangle and asked for the side opposite 30° , just divide by 2 — no other relationship needed.

Q23. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

Since $\sin(\theta) = \text{opposite}/\text{hypotenuse} = 9/41$, consider a right triangle with opposite side 9 and hypotenuse 41.

Find the adjacent side: $(\text{adjacent})^2 + 9^2 = 41^2$, so $(\text{adjacent})^2 = 1681 - 81 = 1600$, giving adjacent = 40.

$\cos(\theta) = \text{adjacent}/\text{hypotenuse} = 40/41$.

WHY EACH OPTION IS WRONG:

- A)** This simply repeats the given sine value rather than computing cosine.
- B)** Correct. The missing leg is 40 (from the 9-40-41 triple), so $\cos(\theta) = 40/41$.
- C)** This subtracts $41 - 9 = 32$ directly instead of using the Pythagorean theorem.
- D)** This compares the two legs to each other ($9/40$), which is $\tan(\theta)$, not $\cos(\theta)$.

TRAP ALERT ⚠️

Taking a shortcut subtraction ($41 - 9$) instead of correctly squaring both values, subtracting, and then taking the square root.

KEY INSIGHT 💡

Never subtract trig ratio numbers directly — the Pythagorean theorem always requires squaring first.

FAST METHOD ⚡

Recognize the 9-40-41 Pythagorean triple immediately once you see 9 and 41 paired as a leg and hypotenuse.

Q24. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

The two acute angles of a right triangle are complementary, meaning they sum to 90° .

Set up the equation: $(3x - 6) + (2x + 16) = 90$.

Combine like terms: $5x + 10 = 90$.

Subtract 10 from both sides: $5x = 80$.

Divide by 5: $x = 16$.

WHY EACH OPTION IS WRONG:

- A)** This comes from incorrectly solving $5x = 90$ (forgetting to subtract the constant +10 first), giving $x = 18$.
- B)** This comes from a sign error, solving as if the constant were -10 instead of $+10$: $5x = 100$, giving $x = 20$.
- C)** Correct. $5x + 10 = 90$ leads to $5x = 80$, so $x = 16$.

D) This comes from setting the two expressions equal to EACH OTHER ($3x - 6 = 2x + 16$) instead of summing them to 90, giving $x = 22$.

TRAP ALERT 🚫

Setting the two angle expressions equal to each other instead of summing them to 90 — a natural instinct when two algebraic expressions appear together, but incorrect here since these are two DIFFERENT angles that must add to a right angle's complement, not be equal to each other.

KEY INSIGHT 💡

Whenever two acute angles of a right triangle are described algebraically, their sum (not their equality) gives you the equation: $\text{angle}_1 + \text{angle}_2 = 90$.

FAST METHOD ⚡

Combine the two expressions immediately into one equation set equal to 90, then solve normally — don't overthink the setup.

Q25. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Let the shortest side (opposite the 30° angle) have length x . Then the long leg is $x\sqrt{3}$, and the hypotenuse is $2x$.

The perimeter is the sum of all three sides: $x + x\sqrt{3} + 2x = 3x + x\sqrt{3} = x(3 + \sqrt{3})$.

Set this equal to the given perimeter: $x(3 + \sqrt{3}) = 60 + 20\sqrt{3}$.

Factor the right side: $60 + 20\sqrt{3} = 20(3 + \sqrt{3})$.

Since both sides share the factor $(3 + \sqrt{3})$, divide it out: $x = 20$.

WHY EACH OPTION IS WRONG:

A) This comes from a miscalculation while trying to isolate x , perhaps dividing by an incorrect factor.

B) This comes from a similar algebraic slip, possibly dividing the wrong portion of the perimeter expression.

C) This comes from another arithmetic error in factoring or dividing the perimeter expression.

D) Correct. Since $60 + 20\sqrt{3} = 20(3 + \sqrt{3})$, and the perimeter formula is $x(3 + \sqrt{3})$, matching coefficients gives $x = 20$.

TRAP ALERT 🚫

Trying to separate the rational part (60) and the radical part ($20\sqrt{3}$) of the perimeter and solve them as two unrelated equations, instead of recognizing that the entire expression factors cleanly to match the triangle's perimeter formula.

KEY INSIGHT 💡

When a perimeter or other total is given in the form $(\text{number}) + (\text{number})\sqrt{3}$ for a 30-60-90 triangle, look for a common factor that matches the $(3 + \sqrt{3})$ pattern from the perimeter formula $x(3 + \sqrt{3})$.

FAST METHOD ⚡

Divide the rational part of the perimeter by 3 as a quick check: $60/3 = 20$. Confirm by dividing the radical coefficient by 1 (the coefficient of $\sqrt{3}$ in the formula): $20\sqrt{3}/\sqrt{3} = 20$. Both give the same x , confirming the answer.

Q26. Correct Answer: **31.9** • MEDIUM 🖋️ GRID-IN

STEP-BY-STEP SOLUTION:

The horizontal distance (60 feet) is adjacent to the angle of elevation, and the tree's height is opposite it.

$$\tan(28^\circ) = \text{opposite/adjacent} = \text{height}/60.$$

$$\text{height} = 60 \times \tan(28^\circ).$$

Using a calculator, $\tan(28^\circ) \approx 0.53171$, so $\text{height} \approx 60 \times 0.53171 \approx 31.9$ feet.

COMMON WRONG ANSWERS & WHY:

- Gridding 28.2 — using $\sin(28^\circ)$ instead of $\tan(28^\circ)$, incorrectly treating the 60-foot distance as the hypotenuse.
- Gridding 68.0 — dividing 60 by $\tan(28^\circ)$ instead of multiplying, inverting the correct setup.

TRAP ALERT

With a non-special angle like 28° , there's no shortcut ratio to fall back on — the calculator's $\tan$ value must be used directly, and it's easy to accidentally reach for $\sin$ or $\cos$ out of habit.

KEY INSIGHT

For any ground-level angle-of-elevation problem with a known horizontal distance, the formula $\text{height} = \text{distance} \times \tan(\text{angle})$ applies regardless of whether the angle is a "special" angle or not.

FAST METHOD

Type $\tan(28)$ into the calculator first, then multiply by 60 — don't try to estimate or simplify a non-special angle by hand.

Q27. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

First find the hypotenuse of triangle ABC using the Pythagorean theorem: $\sqrt{6^2 + 8^2} = \sqrt{36+64} = \sqrt{100} = 10$.

Since triangle DEF is similar to triangle ABC with a scale factor of 5, every side of DEF is 5 times the corresponding side of ABC.

$$\text{Hypotenuse of DEF} = 5 \times 10 = 50.$$

WHY EACH OPTION IS WRONG:

- A)** Correct. The hypotenuse of ABC is 10, and scaling by 5 gives $5 \times 10 = 50$.
- B)** This reports the original hypotenuse of triangle ABC without applying the scale factor at all.
- C)** This scales the SUM of the two legs ($6 + 8 = 14$) by 5, getting 70, instead of first finding the actual hypotenuse and then scaling it.
- D)** This scales only one leg ($8 \times 5 = 40$) instead of computing and scaling the hypotenuse.

TRAP ALERT

Scaling the wrong quantity — either forgetting to scale at all, or scaling some other combination of the given legs instead of first computing the actual hypotenuse of the original triangle.

KEY INSIGHT

To scale ANY side of a similar triangle, first find that exact side's length in the original triangle, then multiply by the scale factor — there are no shortcuts that skip computing the original side first.

FAST METHOD

Recognize the 6-8-10 Pythagorean triple immediately, then just multiply the hypotenuse (10) by the given scale factor (5) to get 50.

Q28. Correct Answer: **B • MEDIUM**

STEP-BY-STEP SOLUTION:

Since $\tan(\theta) = \text{opposite/adjacent} = 8/15$, consider a right triangle with opposite side 8 and adjacent side 15.

Find the hypotenuse using the Pythagorean theorem: $\text{hypotenuse} = \sqrt{8^2 + 15^2} = \sqrt{64+225} = \sqrt{289} = 17$.

$\sin(\theta) = \text{opposite/hypotenuse} = 8/17$.

WHY EACH OPTION IS WRONG:

- A)** This uses the adjacent side over the hypotenuse (15/17), which is $\cos(\theta)$, not $\sin(\theta)$.
- B)** Correct. With hypotenuse 17 (from the 8-15-17 triple), $\sin(\theta) = 8/17$.
- C)** This simply repeats the given tangent value instead of computing sine.
- D)** This is the reciprocal of the given tangent value, unrelated to sine.

TRAP ALERT 🚧

Given only a tangent value (a ratio of two legs), forgetting that the hypotenuse must first be calculated using the Pythagorean theorem before sine or cosine can be found.

KEY INSIGHT 💡

Tangent only involves the two legs, so when you're given $\tan(\theta)$ and asked for sine or cosine, you must first construct the third side (the hypotenuse) using the Pythagorean theorem.

FAST METHOD ⚡

Recognize the 8-15-17 Pythagorean triple immediately once you see $\tan(\theta) = 8/15$, since 8 and 15 are exactly the two legs of that triple.

Q29. Correct Answer: **C • HARD**

STEP-BY-STEP SOLUTION:

The two acute angles of a right triangle are always complementary, meaning they sum to exactly 90° , regardless of the specific value of x .

Since $(5x - 20)^\circ$ and $(x + 38)^\circ$ are complementary angles, the co-function identity applies directly: $\sin((5x - 20)^\circ) = \cos((x + 38)^\circ)$.

Therefore, $\cos((x + 38)^\circ) = \sin((5x - 20)^\circ) = 0.643$, with no need to ever solve for x .

WHY EACH OPTION IS WRONG:

- A)** This computes $1 - 0.643 = 0.357$, incorrectly treating the relationship like complementary probabilities rather than the co-function identity.
- B)** This computes the reciprocal of 0.643 (≈ 1.555), confusing the co-function relationship with a reciprocal relationship (which would apply to tangent, not sine/cosine).
- C)** Correct. Because the two angles are complementary by definition (any right triangle's acute angles sum to 90°), $\cos((x+38)^\circ) = \sin((5x-20)^\circ) = 0.643$ immediately.
- D)** This is a tempting trap: solving for x algebraically IS possible, but completely unnecessary — the co-function identity gives the answer directly without ever finding x .

TRAP ALERT 🚧

Assuming that because two algebraic angle expressions are involved, you must first solve for x before you can find any trig value — when in fact, the complementary relationship makes solving for x entirely optional busywork.

KEY INSIGHT 💡

Any two acute angles in a right triangle are automatically complementary, no matter how their measures are expressed (numerically or algebraically) — the co-function identity $\sin(A) = \cos(B)$ applies immediately without needing their actual numeric measures.

FAST METHOD ⚡

Check whether two given angle expressions are described as the two acute angles of the SAME right triangle — if so, they're complementary by definition, and $\sin$ of one equals $\cos$ of the other without any algebra.

Q30. Correct Answer: **D** • **HARD**

STEP-BY-STEP SOLUTION:

When the altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are both similar to the original triangle and to each other.

This similarity leads to a key relationship: the altitude is the geometric mean of the two segments it creates on the hypotenuse: $CD^2 = AD \times DB$.

$$CD^2 = 4 \times 9 = 36.$$

$$CD = \sqrt{36} = 6.$$

WHY EACH OPTION IS WRONG:

A) This simply adds the two segments ($4 + 9 = 13$), which gives the length of the full hypotenuse AB , not the altitude CD .

B) This is CD^2 , the value right after multiplying AD and DB — the final square root step was skipped.

C) This averages the two segments ($(4+9)/2 = 6.5$) instead of taking their geometric mean (which requires multiplying, then taking a square root, not adding and dividing by 2).

D) Correct. $CD = \sqrt{AD \times DB} = \sqrt{4 \times 9} = \sqrt{36} = 6$.

TRAP ALERT 🚨

Confusing the geometric mean (multiply then take the square root) with the arithmetic mean (add then divide by 2) — these give different results, and only the geometric mean correctly describes the altitude relationship.

KEY INSIGHT 💡

When an altitude is drawn from the right angle of a right triangle to its hypotenuse, that altitude equals the geometric mean of the two segments it creates: $\text{altitude} = \sqrt{(\text{segment}_1 \times \text{segment}_2)}$.

FAST METHOD ⚡

Multiply the two given hypotenuse segments and take the square root in one motion: $\sqrt{4 \times 9} = \sqrt{36} = 6$.

Q31. Correct Answer: **78.6** • **HARD** ✎ **GRID-IN**

STEP-BY-STEP SOLUTION:

Label the known leg as adjacent = 25. Find the opposite leg using tangent: $\tan(40^\circ) = \text{opposite}/25$, so $\text{opposite} = 25 \times \tan(40^\circ)$.

Using a calculator, $\tan(40^\circ) \approx 0.8391$, so opposite $\approx 25 \times 0.8391 \approx 20.98$.

Find the hypotenuse using cosine: $\cos(40^\circ) = 25/\text{hypotenuse}$, so hypotenuse $= 25/\cos(40^\circ)$.

Using a calculator, $\cos(40^\circ) \approx 0.7660$, so hypotenuse $\approx 25/0.7660 \approx 32.64$.

Perimeter $= 25 + 20.98 + 32.64 \approx 78.6$.

COMMON WRONG ANSWERS & WHY:

- Gridding 45.98 — forgetting to include the hypotenuse, adding only the two legs ($25 + 20.98$).
- Gridding 53.6 — forgetting to include one of the legs in the final sum, such as omitting the adjacent leg itself.

TRAP ALERT

With three separate calculator computations required (one leg via tangent, the hypotenuse via cosine, then a final sum), it's easy to lose track of one piece or forget to add all three sides together at the end.

KEY INSIGHT

Multi-step trig word problems often require finding two unknown sides separately (using two different ratios) before combining them — work through each piece methodically and keep track of what's still needed.

FAST METHOD

Once you have the adjacent leg, get the hypotenuse via $\cos(\text{angle}) = \text{adjacent}/\text{hypotenuse}$ rather than recomputing from scratch with the Pythagorean theorem — it's a single calculator step.

Q32. Correct Answer: **A** • **HARD**

STEP-BY-STEP SOLUTION:

Since $\cos(\theta) = \text{adjacent}/\text{hypotenuse} = 12/37$, consider a right triangle with adjacent side 12 and hypotenuse 37.

Find the opposite side using the Pythagorean theorem: $(\text{opposite})^2 + 12^2 = 37^2$, so $(\text{opposite})^2 = 1369 - 144 = 1225$, giving opposite $= 35$.

$\sin(\theta) = \text{opposite}/\text{hypotenuse} = 35/37$.

WHY EACH OPTION IS WRONG:

- A**) Correct. The missing leg is 35 (from the 12-35-37 triple), so $\sin(\theta) = 35/37$.
- B**) This simply repeats the given cosine value instead of computing sine.
- C**) This subtracts $37 - 12 = 25$ directly instead of using the Pythagorean theorem correctly.
- D**) This compares the two legs to each other ($12/35$), which is $\tan(\theta)$ (or its reciprocal), not $\sin(\theta)$.

TRAP ALERT

The 12-35-37 triple is far less commonly memorized than triples like 3-4-5 or 5-12-13, making it tempting to take a shortcut (like direct subtraction) instead of carefully applying the Pythagorean theorem.

KEY INSIGHT

When a Pythagorean triple isn't immediately recognizable, don't guess — apply the Pythagorean theorem methodically: square both given values, subtract, then take the square root.

FAST METHOD

Even unfamiliar triples are easy to verify with a quick calculation: $12^2 + 35^2 = 144 + 1225 = 1369 = 37^2$, confirming the triple before trusting the answer.

Q33. Correct Answer: **B • HARD**

STEP-BY-STEP SOLUTION:

The angle the ladder makes with the GROUND is opposite the height it reaches up the wall (18 feet) and has the ladder itself (20 feet) as the hypotenuse.

$\sin(\text{angle}) = \text{opposite/hypotenuse} = 18/20 = 0.9$.

To find the angle itself, use the inverse sine function: $\text{angle} = \sin^{-1}(0.9)$.

Using a calculator, $\sin^{-1}(0.9) \approx 64.16^\circ$, which rounds to 64° .

WHY EACH OPTION IS WRONG:

A) This results from using $\cos^{-1}(18/20)$ instead of $\sin^{-1}(18/20)$, incorrectly treating the height as adjacent to the ground angle instead of opposite to it.

B) Correct. The angle with the ground satisfies $\sin(\text{angle}) = 18/20$, so $\text{angle} = \sin^{-1}(0.9) \approx 64^\circ$.

C) This results from a different miscalculation, such as using an incorrect ratio setup entirely.

D) This results from yet another plausible but incorrect ratio setup.

TRAP ALERT 🚧

Confusing which side is opposite the angle at the BASE of the ladder — the height up the wall is opposite this angle (not adjacent), since the wall and the angle at the base are on opposite sides of the triangle.

KEY INSIGHT 💡

For a ladder problem, the angle with the GROUND has the height as its opposite side and the base distance as its adjacent side; the angle with the WALL has these two sides reversed.

FAST METHOD ⚡

Picture the triangle: the angle at the base looks UP at the wall, so the height (the far side) must be opposite, and sine is the right tool when the hypotenuse (ladder) is also known.

Q34. Correct Answer: **C • HARD**

STEP-BY-STEP SOLUTION:

Apply the Pythagorean theorem: $(3x)^2 + (4x)^2 = 20^2$.

Expand: $9x^2 + 16x^2 = 400$.

Combine like terms: $25x^2 = 400$.

Divide both sides by 25: $x^2 = 16$.

Take the square root: $x = 4$ (using the positive root, since x represents a length scale factor).

WHY EACH OPTION IS WRONG:

A) This comes from an arithmetic slip while solving $25x^2 = 400$, such as dividing incorrectly.

B) This comes from a similar miscalculation in isolating x .

C) Correct. $25x^2 = 400$ leads to $x^2 = 16$, so $x = 4$.

D) This comes from yet another arithmetic slip, possibly dividing 400 by the wrong coefficient.

TRAP ALERT 🚧

Recognizing that $3x$ and $4x$ resemble a 3-4-5 triangle pattern is helpful, but it's still essential to carry out the full algebra (squaring, combining, dividing, and taking a square root) rather than guessing the scale factor.

KEY INSIGHT 💡

When a right triangle's legs are given as multiples of a variable (like $3x$ and $4x$), the Pythagorean theorem becomes a one-variable quadratic equation that solves cleanly when the legs follow a recognizable ratio.

FAST METHOD ⚡

Since $3x$ and $4x$ mirror the 3-4-5 ratio, the hypotenuse should be $5x$. Setting $5x = 20$ immediately gives $x = 4$, skipping the full Pythagorean expansion.

Q35. Correct Answer: **25.6** • **HARD** 🖊️ **GRID-IN**

STEP-BY-STEP SOLUTION:

Since the angle is measured between the wire and the pole (a vertical segment), the pole (21 feet) is adjacent to this angle, and the wire is the hypotenuse of the right triangle.

$\cos(35^\circ) = \text{adjacent/hypotenuse} = 21/\text{wire length}$.

$\text{wire length} = 21/\cos(35^\circ)$.

Using a calculator, $\cos(35^\circ) \approx 0.8192$, so $\text{wire length} \approx 21/0.8192 \approx 25.6$ feet.

COMMON WRONG ANSWERS & WHY:

- Gridding 36.6 — using $\sin(35^\circ)$ instead of $\cos(35^\circ)$, incorrectly treating the pole as opposite the given angle instead of adjacent to it.
- Gridding 14.7 — multiplying 21 by $\cos(35^\circ)$ instead of dividing, inverting the correct relationship.

TRAP ALERT 🚧

The angle is measured from the POLE, not the ground as in most angle-of-elevation problems — this swaps which leg is adjacent and which is opposite compared to the more familiar setup, and it's easy to apply the usual ground-angle formula out of habit.

KEY INSIGHT 💡

Always identify exactly which two sides the given angle sits between — an angle measured from a vertical pole makes that pole the ADJACENT side, not the opposite side, flipping the usual tangent-based setup into a cosine-based one.

FAST METHOD ⚡

When an angle is described as being with a specific segment (here, "with the pole"), that segment is always adjacent to the angle — read word problems carefully to identify this before choosing $\sin$, $\cos$, or $\tan$.

Q36. Correct Answer: **D** • **HARD**

STEP-BY-STEP SOLUTION:

Since angle A and angle B are complementary (they sum to 90°), $\tan(B) = \tan(90^\circ - A)$.

The identity $\tan(90^\circ - A) = 1/\tan(A)$ holds for any acute angle A (tangent and its complementary partner are RECIPROCALs, unlike sine and cosine, which are EQUAL).

$\tan(B) = 1/\tan(A) = 1/(5/12) = 12/5$.

WHY EACH OPTION IS WRONG:

A) This incorrectly applies the sine/cosine equality rule ($\sin(A)=\cos(B)$) to tangent, assuming $\tan(B)$ simply equals $\tan(A)$ — but complementary tangents are reciprocals, not equal.

B) This computes a side-ratio value ($5/13$, using a 5-12-13 triangle) that doesn't correspond to $\tan(B)$ at all.

- C)** This computes another side-ratio value ($12/13$) that also doesn't correspond to $\tan(B)$.
D) Correct. Since complementary angles have reciprocal tangents, $\tan(B) = 1/\tan(A) = 12/5$.

TRAP ALERT 🚫

Applying the familiar $\sin(A) = \cos(B)$ co-function rule directly to tangent, assuming by analogy that $\tan(A) = \tan(B)$ for complementary angles — but tangent behaves differently: $\tan(A)$ and $\tan(B)$ are reciprocals of each other, not equal.

KEY INSIGHT 💡

For complementary angles A and B : $\sin(A) = \cos(B)$ and $\cos(A) = \sin(B)$ (equal pairs), but $\tan(A) = 1/\tan(B)$ (reciprocal pair) — tangent is the exception to the "just copy the value" shortcut.

FAST METHOD ⚡

If a question asks about $\tan(B)$ where B is complementary to A , immediately flip $\tan(A)$ upside down — never copy it directly the way you would with $\sin/\cos$.

Q37. Correct Answer: **A** • **HARD**

STEP-BY-STEP SOLUTION:

The pole, the ground distance, and the wire form a right triangle, with the wire as the hypotenuse (it's the longest side, stretching from the top of the pole to a point on the ground).

Apply the Pythagorean theorem: $16^2 + 30^2 = (\text{wire})^2$.

$$256 + 900 = 1156.$$

$$\text{wire} = \sqrt{1156} = 34 \text{ feet.}$$

WHY EACH OPTION IS WRONG:

- A)** Correct. $\sqrt{16^2 + 30^2} = \sqrt{1156} = 34$ feet.
B) This adds the two given lengths directly ($16 + 30 = 46$) instead of applying the Pythagorean theorem.
C) This subtracts the two given lengths directly ($30 - 16 = 14$), again ignoring the actual relationship between the sides.
D) This subtracts the squares instead of adding them ($\sqrt{30^2 - 16^2} \approx 25.4$), which would only be appropriate if solving for a LEG rather than the hypotenuse.

TRAP ALERT 🚫

Subtracting the squares instead of adding them, a mistake that happens when a student misremembers which configuration (solving for a leg vs. solving for the hypotenuse) calls for addition versus subtraction.

KEY INSIGHT 💡

Whenever the unknown side is the LONGEST side of the triangle (connecting two of the farthest-apart points, like the top of a pole to a distant ground point), it's the hypotenuse, and the Pythagorean theorem requires ADDING the squares of the two known legs.

FAST METHOD ⚡

Recognize the 16-30-34 Pythagorean triple (twice the 8-15-17 triple) once you see 16 and 30 paired as the two legs.

Q38. Correct Answer: **B** • **HARD**

STEP-BY-STEP SOLUTION:

First find the hypotenuse of triangle ABC: $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

Since triangle DEF is similar to triangle ABC, find the scale factor by comparing the hypotenuses:
scale factor = $25/10 = 2.5$.

Apply this same scale factor to the leg of length 6: corresponding leg in DEF = $6 \times 2.5 = 15$.

WHY EACH OPTION IS WRONG:

- A)** This is the OTHER leg's corresponding value ($8 \times 2.5 = 20$) — the scale factor was applied correctly, but to the wrong original leg.
- B)** Correct. Scale factor = $25/10 = 2.5$, so the corresponding leg is $6 \times 2.5 = 15$.
- C)** This results from incorrectly using half of the hypotenuse ($25/2 = 12.5$) as if it were the answer directly.
- D)** This results from scaling using the wrong original side as the basis for the scale factor: $6 \times (25/8) = 18.75$, mistakenly using the OTHER leg (8) instead of the hypotenuse (10) to find the scale factor.

TRAP ALERT

Using the wrong pair of corresponding sides to establish the scale factor — the scale factor must come from comparing two sides that are KNOWN in both triangles (here, the two hypotenuses), not from guessing based on whichever numbers are available.

KEY INSIGHT

To scale between similar triangles, always establish the scale factor using a pair of corresponding sides where BOTH values are known, then apply that single factor consistently to every other side.

FAST METHOD

Find the original triangle's hypotenuse first (here, the 6-8-10 triple gives hypotenuse 10), then divide the new hypotenuse by the old one for your scale factor: $25/10 = 2.5$.

Q39. Correct Answer: **132** • **HARD**  **GRID-IN**

STEP-BY-STEP SOLUTION:

The string is the hypotenuse of a right triangle, the angle with the ground is 62° , and the kite's height is opposite this angle.

$$\sin(62^\circ) = \text{opposite/hypotenuse} = \text{height}/150.$$

$$\text{height} = 150 \times \sin(62^\circ).$$

Using a calculator, $\sin(62^\circ) \approx 0.8829$, so height $\approx 150 \times 0.8829 \approx 132.4$, which rounds to 132 feet.

COMMON WRONG ANSWERS & WHY:

- Gridding 70 — using $\cos(62^\circ)$ instead of $\sin(62^\circ)$, incorrectly computing the horizontal distance to the kite instead of its height.
- Gridding 170 — dividing 150 by $\sin(62^\circ)$ instead of multiplying, inverting the correct setup.

TRAP ALERT

The angle is given with the ground, so it's tempting to immediately reach for tangent (as in many elevation problems), but here the HYPOTENUSE (string length) is known, not a leg — making sine the correct tool instead.

KEY INSIGHT

Identify which two pieces of information you're actually given before choosing sin, cos, or tan: when the hypotenuse and an angle are known and you need the opposite side, sine is the correct ratio (not tangent, which needs two legs).

FAST METHOD

Whenever a string, wire, ladder, or ramp LENGTH is given directly (not a horizontal distance), that length is the hypotenuse — default to sine or cosine, not tangent.

Q40. Correct Answer: **C** • **HARD**

STEP-BY-STEP SOLUTION:

Let the sides be $3k$, $4k$, and $5k$ for some positive value k . Since $3^2 + 4^2 = 5^2$, this ratio always forms a valid right triangle, with the two smaller terms ($3k$ and $4k$) as the legs and $5k$ as the hypotenuse.

The perimeter is the sum of all three sides: $3k + 4k + 5k = 12k$.

Set this equal to the given perimeter: $12k = 60$, so $k = 5$.

The legs are $3k = 15$ and $4k = 20$ (the hypotenuse, $5k = 25$, is not used for area since it's not one of the two legs).

Area = $(1/2)(15)(20) = 150$.

WHY EACH OPTION IS WRONG:

A) This comes from an arithmetic slip while computing half the product of the legs.

B) This is the full product of the legs ($15 \times 20 = 300$) without applying the necessary factor of one-half.

C) Correct. With legs 15 and 20, area = $(1/2)(15)(20) = 150$.

D) This mistakenly uses the hypotenuse (25) as if it were a leg, computing $(1/2)(15)(25) = 187.5$ instead of using the two actual legs.

TRAP ALERT 🚧

After finding all three side lengths, mistakenly using the hypotenuse as one of the two sides in the area formula instead of recognizing that area only ever uses the two legs (the sides that form the right angle).

KEY INSIGHT 💡

For a right triangle, area = $(1/2)(\text{leg}_1)(\text{leg}_2)$ always uses the two legs that meet at the right angle — never the hypotenuse, since the hypotenuse is not perpendicular to either leg.

FAST METHOD ⚡

In a 3:4:5 ratio, the two SMALLER numbers (3 and 4) are always the legs, and the largest (5) is always the hypotenuse — use only the two smaller scaled values for area.

TOPIC SUMMARY — Right Triangle Trigonometry

Common Mistakes:

1. Mixing up opposite and adjacent: the opposite side never touches the angle in question, while the adjacent leg always does (besides the hypotenuse). Pulling the hypotenuse into a tangent ratio, or leaving it out of a sine/cosine ratio, is the most common slip.
2. Treating the tangent co-function relationship the same as sine/cosine: for complementary angles A and B , $\sin(A) = \cos(B)$ and $\cos(A) = \sin(B)$ (equal), but $\tan(A) = 1/\tan(B)$ (reciprocal, NOT equal).
3. Doing unnecessary work — when two angles in the same right triangle are described as complementary (numerically or algebraically), sin of one equals cos of the other immediately, with no need to solve for an unknown angle or rebuild the triangle from scratch.

Must Remember:

1. SOH-CAH-TOA: $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$. The two acute angles of any right triangle always sum to 90° (they're complementary).
2. Special right triangles: 45-45-90 has sides in ratio $1 : 1 : \sqrt{2}$; 30-60-90 has sides in ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). For any right triangle, the Pythagorean theorem $a^2 + b^2 = c^2$ connects the two legs to the hypotenuse.
3. Co-function identity: $\sin(\theta^\circ) = \cos((90-\theta)^\circ)$ for any acute angle. Word problems use $\tan$ when the horizontal distance (adjacent) and height (opposite) are involved, but $\sin$ or $\cos$ when a wire, ladder, or ramp LENGTH (the hypotenuse) is given directly.

Expected on DSAT:

Based on this project's frequency analysis of past Digital SAT papers, right triangle trigonometry appeared in 11 of 1,240 math questions analyzed (about 0.9%). That works out to roughly 0–1 question per test on average, appearing in only about 1 of every 3 tests — but the co-function (complementary angle) trap shows up disproportionately often within that small sample, making it the single highest-value pattern to master for this topic.

— END OF SOLUTIONS —